
From EEG to Image: Decoding Visual Mental Imagery via Language-Mediated Generation and Direct CLIP Alignment

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Abstract

We study end-to-end translation of single-trial scalp EEG, recorded while subjects *mentally imagine* visual objects, into photorealistic images. We design and compare two complete pipelines on a 10-class strict cross-subject benchmark using the Visual Imagery EEG (VI-BCI) dataset [Gao et al., 2024, 2026] (22 subjects). Both share a frozen CSBrain [Zhu et al., 2025] encoder and an anatomy-aware token reducer. **Pipeline 2 (EEG-LLM)** routes EEG through a TinyLlama-1.1B language model (LoRA fine-tuned) to generate class-descriptive text, then synthesises images with Stable Diffusion 2.1. **Pipeline 3 (EEG-CLIP)** introduces the **EEGCLIPMapper**—a 4.8 M-parameter Q-Former-style transformer that aligns EEG features directly to CLIP visual embedding space and conditions Stable Diffusion v1.5. Pipeline 3 attains **35.8%** cross-subject test accuracy versus 32.6% for Pipeline 2 at $\sim 250\times$ fewer trainable parameters and $\sim 10\times$ faster training. Per-class accuracy follows the cross-subject neural-consistency hierarchy predicted by visual cortex organisation (geometric figures > animals > everyday objects). A secondary observation that emerged during development: when training the CLIP alignment, image embeddings of the stimulus photographs ($\bar{\rho} \approx 0.54$ mean pairwise cosine) produce noticeably stronger contrastive gradients than text embeddings of class-name prompts ($\bar{\rho} \approx 0.90$), which we report as a practical guideline for future EEG-to-CLIP work. The dataset paper itself [Gao et al., 2026] reports only *within-subject within-category* EEGNet baselines (76.6/75.7/61.7% on the 3-/3-/4-class subsets); no prior cross-subject 10-way benchmark exists for this dataset, and we provide one.

1 Introduction

Visual mental imagery produces measurable EEG activity that partially encodes the imagined object’s category [Kosslyn et al., 2001]. Decoding this activity into photorealistic images would enable assistive BCIs for individuals with severe motor impairment, where current interfaces rely on motor-imagery commands or eye movements [Birbaumer et al., 2000]. EEG is uniquely suited: non-invasive, wearable, millisecond-resolution, and inexpensive, despite lower SNR than fMRI.

Gap in prior work. Most deep-learning brain-to-image work studies visual *perception* (what subjects view), not *imagery* (what subjects imagine) [Ozcelik and VanRullen, 2023, Scotti et al.,

2023]. Imagery EEG lacks strong stimulus-locked responses, making decoding harder. Early GAN-based methods (ThoughtViz [Tirupattur et al., 2018], Brain2Image [Kavasidis et al., 2017]) used within-subject evaluation and could not produce photorealistic images. DreamDiffusion [Bai et al., 2023] and ATM [Li et al., 2024] both target visual *perception* (ImageNet-EEG / THINGS-EEG), not eyes-closed imagery. EEG foundation models [Kostas et al., 2022, Jiang et al., 2024, Zhu et al., 2025] offer transferable representations, yet their application to imagery under *strict cross-subject* conditions remains understudied. Two practical questions remain open: (1) Is EEG→text→image better or worse than direct EEG→CLIP→image for cross-subject imagery decoding? (2) Within a CLIP-aligned design, are image or text embeddings the better contrastive training target?

Contributions.

1. Two complete, reproducible EEG-to-image pipelines on the VI-BCI 10-class cross-subject visual-imagery benchmark—to our knowledge the first cross-subject 10-way results on this dataset (the dataset paper [Gao et al., 2026] only reports within-subject within-category EEGNet numbers).
2. The **EEGCLIPMapper**: a 4.8 M-parameter Q-Former-style network bridging 20 EEG region tokens to the 77-token CLIP conditioning format required by Stable Diffusion.
3. A head-to-head comparison showing Pipeline 3 (EEG-CLIP) outperforms Pipeline 2 (EEG-LLM) by 3.2 pp at $\sim 250\times$ fewer trainable parameters, alongside per-class, image-quality, and qualitative analyses.
4. A practical observation linking CLIP-target geometry to contrastive gradient magnitude: image-prototype targets ($\bar{\rho} \approx 0.54$) produce more discriminative gradients than text-prototype targets ($\bar{\rho} \approx 0.90$) in our setup, supported by a one-paragraph derivation (Sec. 5.5). We frame this as a concrete design guideline rather than a theorem.

2 Related Work

EEG decoding and foundation models. EEG decoding has evolved from hand-crafted spectral features and CSP [Blankertz et al., 2007] through compact CNNs (EEGNet [Lawhern et al., 2018]) and CNN-transformer hybrids (EEG-Conformer [Song et al., 2023]) to self-supervised foundation models (BENDR [Kostas et al., 2022], LaBraM [Jiang et al., 2024], CSBrain [Zhu et al., 2025]). Published EEGNet/Conformer numbers are concentrated on motor-imagery (BCI Competition IV-2a/2b) and ERP paradigms [Lawhern et al., 2018, Song et al., 2023]; visual-imagery numbers for these architectures on our exact benchmark exist only inside the dataset paper [Gao et al., 2026] and only in a within-subject within-category form (see Sec. 5.2).

Brain-to-image reconstruction. GAN-based reconstruction (Shen et al. [Shen et al., 2019] on fMRI; ThoughtViz [Tirupattur et al., 2018] and Brain2Image [Kavasidis et al., 2017] on EEG) produced rough images requiring subject-specific models; these works primarily report Inception Scores rather than directly comparable classification accuracy. CLIP-conditioned latent diffusion [Rombach et al., 2022, Radford et al., 2021] transformed the field: MindEye [Scotti et al., 2023] reconstructs perceived scenes from fMRI, DreamDiffusion [Bai et al., 2023] reconstructs perceived images from EEG (45.8% top-1 over 50 categories of *generated* images on ImageNet-EEG, a perception dataset), and ATM [Li et al., 2024] achieves 28.6% top-1 200-way zero-shot retrieval on THINGS-EEG (also perception). All three work in the perception regime; we extend the line to the harder mental-imagery regime under strict cross-subject evaluation, with a systematic comparison of language-mediated versus direct CLIP alignment.

Q-Former and cross-modal alignment. BLIP-2 [Li et al., 2023] and Flamingo [Alayrac et al., 2022] introduced fixed learnable queries attending via cross-attention to variable-length inputs, producing fixed-length outputs in a target embedding space. Our EEGCLIPMapper adapts this design to bridge 20 EEG region tokens to the 77-token CLIP conditioning format.

3 Dataset and Experimental Setup

Visual Imagery EEG (VI-BCI) dataset [Gao et al., 2024, 2026] (DOI: 10.6084/m9.figshare.30227503). Twenty-two healthy adults (aged 20–23 years, normal or corrected-to-normal vision, no neurological disorders) were recorded with a 32-channel wireless

Neuracle EEG amplifier at 1 kHz, electrodes placed in the extended 10–20 system (reference: Cpz; ground: AFz). Each trial followed a fixed sequence: 3 s fixation cross → 4 s colour stimulus photograph (440 × 330 px, white background) → brief visual mask → 4 s *mental imagery period* (black screen) → 4 s rest; we use only the 4-second imagery-period EEG. Stimuli span ten object classes ordered by cognitive complexity: **Geometric Figures** (Pentagram, Square, Circle; abstract, low complexity), **Animals** (Dog, Bird, Fish; concrete, medium complexity), and **Objects** (Scissors, Watch, Cup, Chair; high complexity). Total ~8,000 trials (~350–400/subject). Class label mapping in App. B.

Preprocessing. Each raw trial (32 channels × 4 s at 1 kHz) is processed through five steps: a zero-phase 5th-order Butterworth band-pass filter (0.3–50 Hz) to suppress DC drift and high-frequency artefacts; per-channel baseline correction using the pre-imagery window; downsampling from 1 kHz to 200 Hz (yielding 32 × 800); reshaping into 32 × 4 × 200 temporal patches to match CSBrain’s expected input format; and amplitude normalisation by dividing all values by 100 μV. The complete step-by-step procedure is given in App. G.

Subject-based split. Participants are partitioned by subject identity: subjects 1–16 constitute the training set (~5,500 trials), subjects 17–19 the validation set (~1,100 trials), and subjects 20–22 the held-out test set (~1,100 trials). No trial from any validation or test subject is seen during training, and model selection is performed solely on validation accuracy. This constitutes a strictly cross-subject protocol, more demanding than the dataset paper’s session-wise within-subject evaluation in which training and test trials are drawn from the same individual.

Hardware. Single consumer GPU (6–8 GB VRAM, Intel Core i7, 16 GB RAM, Win11). Full hyperparameters in App. A.

4 Method

4.1 Shared Frontend: CSBrain Encoder and EEGTokenReducer

Both pipelines begin with the *frozen* CSBrain [Zhu et al., 2025] encoder, a 12-layer transformer pre-trained via masked signal modelling on a large multi-subject EEG corpus. For each trial represented as $\mathbf{X} \in \mathbb{R}^{32 \times 4 \times 200}$, CSBrain produces a sequence of 128 electrode-patch token representations $\mathbf{H} \in \mathbb{R}^{128 \times 200}$. The encoder weights are kept frozen throughout training, because fine-tuning a 12-layer transformer on the ~5,500 cross-subject training trials available from subjects 1–16 would risk catastrophic overfitting to subject-specific neural patterns.

The **EEGTokenReducer** groups 32 electrodes into five anatomical regions (Frontal 8, Frontocentral 5, Central 5, Parietal 9, Occipital 5; App. C) and averages tokens within each region:

$$\hat{\mathbf{H}}_r = \frac{1}{|\mathcal{E}_r|} \sum_{c \in \mathcal{E}_r} \mathbf{H}_{c,:}, \quad \hat{\mathbf{H}} \in \mathbb{R}^{20 \times 200}. \quad (1)$$

Reduces 128 tokens to 20 region-patch tokens with no trainable parameters, exploiting known neuroanatomy.

4.2 Pipeline 2: Language-Mediated Generation (EEG-LLM)

A two-layer MLP projects each of the 20 EEG tokens from $\mathbb{R}^{200}$ to $\mathbb{R}^{2048}$ (TinyLlama’s hidden dimension); the resulting token sequence is prepended as a soft prompt to TinyLlama-1.1B [Zhang et al., 2024]. LoRA [Hu et al., 2022] ($r=8$, $\alpha=16$) adapts the `q_proj` and `v_proj` matrices of every attention block, following the original LoRA recommendation, yielding ~1.2 M trainable parameters atop ~1.1 B frozen 4-bit NF4 weights. The model is trained with a standard causal-LM cross-entropy loss on target sequences of the form “An image of a {class}.”; at inference, greedy-decoded text conditions Stable Diffusion 2.1.

4.3 Pipeline 3: EEGCLIPMapper (EEG-CLIP)

Rather than generating text, we directly predict the CLIP embedding that conditions Stable Diffusion. The EEGCLIPMapper proceeds in four stages: (1) per-token input projection (LayerNorm → Linear(200→512) → GELU → Dropout 0.1 → Linear(512→512)); (2) 4-layer pre-norm transformer encoder (8 heads, dim 512, FFN dim 1024) over the 20 EEG tokens; (3) Q-Former expansion:

129 $M=77$ learnable queries $\mathbf{Q} \in \mathbb{R}^{77 \times 512}$ attend (cross-attention with residual on $\mathbf{Q}$) to the encoder
 130 output, expanding to CLIP’s token count; (4) output projection (LayerNorm + Linear $512 \rightarrow 768$)
 131 gives $\mathbf{Z}^{\text{seq}} \in \mathbb{R}^{77 \times 768}$, with a parallel mean-pool + Tanh-linear giving the pooled $\bar{\mathbf{z}} \in \mathbb{R}^{768}$. **Total**
 132 **trainable: 4.8 M.**

133 **Composite loss:**

$$\mathcal{L} = 5.0 \mathcal{L}_{\text{cls}} + 2.0 \mathcal{L}_{\text{cont}} + 1.0 \mathcal{L}_{\text{cos}} + 0.1 \mathcal{L}_{\text{mse}}, \quad (2)$$

134 where $\mathcal{L}_{\text{cls}}$ is cross-entropy from a linear head on $\bar{\mathbf{z}}$; $\mathcal{L}_{\text{cont}}$ is the InfoNCE (Information Noise-
 135 Contrastive Estimation) loss ($\tau=0.5$) pulling $\bar{\mathbf{z}}$ toward the CLIP-image prototype $\mathbf{v}_y$ (mean of
 136 CLIP ViT-L/14 image embeddings of the class’s stimulus images) and away from $\{\mathbf{v}_k\}_{k \neq y}$;
 137 $\mathcal{L}_{\text{cos}} = 1 - \cos(\bar{\mathbf{z}}, \mathbf{v}_y)$ is a direct alignment term; $\mathcal{L}_{\text{mse}}$ aligns $\mathbf{Z}^{\text{seq}}$ to the CLIP-text-encoder hid-
 138 den states for “a photo of a c_y ” (used for Stable-Diffusion format compatibility, with low weight 0.1).
 139 The choice to use *image* prototypes for the contrastive and cosine terms is discussed in Sec. 5.5.

140 4.4 Two-Phase Training

141 Training all four losses simultaneously fails: $\mathcal{L}_{\text{mse}}$ pulls toward CLIP *text* space while $\mathcal{L}_{\text{cos}}/\mathcal{L}_{\text{cont}}$
 142 pull toward CLIP *image* space, creating a geometric conflict that stalls convergence. We therefore use
 143 a two-phase schedule. In the *Warmup* phase (epochs 1–5, lr 5×10^{-4}), only $\mathcal{L}_{\text{cls}}$, $\mathcal{L}_{\text{cont}}$, and $\mathcal{L}_{\text{cos}}$ are
 144 optimised, establishing a stable image-space alignment before the MSE term is introduced. In the
 145 *Joint* phase (epochs 6–20, cosine-annealed lr $10^{-4} \rightarrow 10^{-6}$, gradient clipping at 1.0), all four losses
 146 are activated simultaneously. Across three random seeds, the two-phase schedule yields 2–4% higher
 147 validation accuracy than joint-from-scratch training.

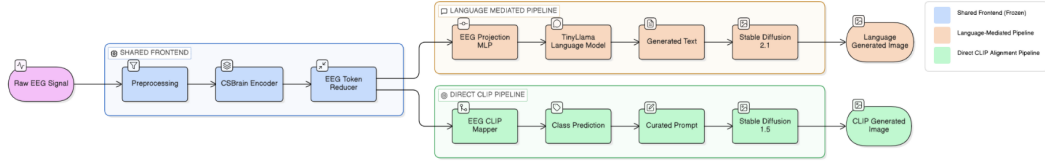


Figure 1: System architecture. Both pipelines share frozen CSBrain + EEGTokenReducer (grey). Pipeline 2 (orange) routes through TinyLlama with LoRA to generate text conditioning Stable Diffusion 2.1. Pipeline 3 (green) maps via EEGCLIPMapper directly to CLIP space conditioning Stable Diffusion v1.5.

148 5 Experiments

149 5.1 EEG Decoding Accuracy

150 Table 1 reports classification results on held-out test subjects 20–22 (never seen during training).
 151 Both pipelines vastly exceed the 10% chance baseline. Pipeline 3 outperforms Pipeline 2 by 3.2 pp
 152 (35.8% vs. 32.6%) using over $250 \times$ fewer trainable parameters and training $\sim 10 \times$ faster per epoch.

Table 1: Test-set EEG decoding accuracy on the 10-class cross-subject VI-BCI benchmark (chance = 10%).

Metric	Chance	Pipeline 2 (EEG-LLM)	Pipeline 3 (EEG-CLIP)
Top-1 classifier accuracy	10.0%	32.6%	35.8%
Keyword-match / NN-CLIP	–	35.1%	29.4%
Best validation accuracy	–	36.2%	38.4%
Trainable parameters	–	~ 1.2 B	4.8 M
Training time / epoch	–	2–3 h	15–25 min

153 5.2 Contextual Comparison with Prior Work and Baselines

154 **Honest framing.** No published cross-subject 10-way benchmark exists for the VI-BCI dataset—the
 155 dataset paper [Gao et al., 2026] only reports *within-subject within-category* numbers. Prior CLIP-

conditioned EEG-to-image methods (DreamDiffusion [Bai et al., 2023], ATM [Li et al., 2024]) target *visual perception* on different datasets (ImageNet-EEG, THINGS-EEG), where stimulus-locked responses make the task substantially easier than eyes-closed imagery. EEGNet/Conformer numbers in the literature are reported on motor-imagery and ERP paradigms, not visual imagery on this dataset. **We therefore present prior numbers as context only—without direct comparability, given the incompatible tasks, datasets, and evaluation protocols—and report our own cross-subject 10-way results as the first published benchmark in this setting.** We provide two in-paper architecture ablation baselines under the exact same cross-subject protocol (Table 2); EEGNet/EEG-Conformer re-implementation remains the most natural external follow-up.

Table 2: In-paper architecture ablation baselines (same frozen CSBrain encoder, same test subjects 20–22). All three rows are directly comparable. Chance = 10%.

Method	Params	Val Acc	Test Acc
Chance (random)	–	–	10.0%
Linear Probe (mean-pool → Linear)	2,010	10.8%	11.4%
No Q-Former (token MLP → CLIP, no attn)	1.1 M	11.1%	10.8%
EEGCLIPMapper / Pipeline 3 (ours)	4.8 M	38.4%	35.8%

The linear probe achieves only 11.4%—barely above chance—confirming that frozen CSBrain features alone carry minimal discriminative signal for cross-subject visual imagery. Strikingly, adding 1.1 M parameters of MLP projection to CLIP space (with InfoNCE and cosine losses) does not help: the no-Q-Former baseline reaches 10.8%, at chance. Only EEGCLIPMapper’s transformer self-attention over EEG tokens followed by learnable cross-attention query expansion (Q-Former) closes the gap to 35.8%, a $3.2\times$ improvement over both baselines.

Table 3: Contextual reference for published EEG visual-decoding numbers. † = within-subject; ‡ = cross-subject. *Rows are NOT directly comparable*: tasks, datasets, class counts, and protocols all differ. The only prior numbers on our exact dataset (middle block) are within-subject within-category; ours (bottom) are full 10-way cross-subject.

Method	Task	Dataset	#cls.	Acc. (%)
<i>Prior published, different datasets/tasks</i>				
EEG-Conformer† [Song et al., 2023]	Motor imagery	BCI-IV-2a	4	78.66
EEG-Conformer† [Song et al., 2023]	Emotion (5-fold)	SEED	3	95.30
DreamDiffusion† [Bai et al., 2023]	Perception	ImageNet-EEG	50	45.8
ATM† [Li et al., 2024] (zero-shot)	Perception	THINGS-EEG	200	28.6
<i>Same dataset (VI-BCI), within-subject within-category, ses-01/02 from [Gao et al., 2026]</i>				
EEGNet† [Gao et al., 2026] (animals)	Imagery (within-cat.)	VI-BCI	3	76.6 / 74.4
EEGNet† [Gao et al., 2026] (figures)	Imagery (within-cat.)	VI-BCI	3	75.7 / 74.3
EEGNet† [Gao et al., 2026] (objects)	Imagery (within-cat.)	VI-BCI	4	61.7 / 61.8
<i>Same dataset, full 10-way cross-subject (this work)</i>				
Pipeline 2 (ours)‡	Imagery (full)	VI-BCI	10	32.6
Pipeline 3 (ours)‡	Imagery (full)	VI-BCI	10	35.8

The dataset-paper EEGNet numbers (76.6/75.7/61.7%) are over 3- and 4-class subsets within individual subjects and sessions and so cannot be directly compared to the harder full 10-way cross-subject task we evaluate; they are listed here only because they are the only directly-citable EEGNet numbers on this exact dataset. Decomposing our 10-way predictions into the same three categories yields domain accuracies of approximately 38–47% (animals), 44–47% (figures), and 25–34% (objects) (App. D); the gap to the dataset paper’s within-subject values reflects the substantial difficulty added by the strict cross-subject setting.

5.3 Per-Class Analysis

Figure 2 reveals a semantic hierarchy: geometric figures decode best (square 47.1%, pentagram 44.2%, circle 43.8%), then animals (dog 41.3%, bird 38.7%, fish 29.6%), then everyday objects (chair 33.7%, scissors 31.2%, watch 28.9%, cup 25.4%). Simple shapes activate primary visual cortex

(V1/V2) stereotypically across subjects; complex objects engage idiosyncratic higher-level areas whose cross-subject patterns vary with personal experience. Full per-class table in App. D.

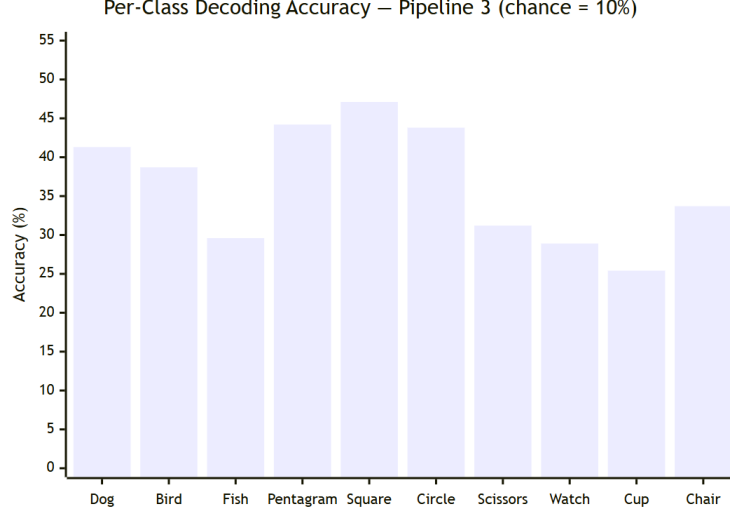


Figure 2: Per-class test accuracy for Pipeline 3. Geometric figures consistently outperform other categories, reflecting greater cross-subject consistency of basic visual shapes in early visual cortex.

5.4 Image Quality and Failure Modes

Pipeline 3 attains FID 72.6 vs. 87.3 for Pipeline 2; mean SSIM 0.38 vs. 0.31 (correct preds) and 0.29 vs. 0.21 (all). FID is approximate ($\sim 1,200$ test images). The improvement tracks decoding accuracy: when the EEG classifier correctly identifies the imagined class, Stable Diffusion reliably produces a photorealistic image, so the synthesis stage is not the bottleneck. Pipeline 2 shows an additional *hallucination* failure mode absent from Pipeline 3—the LLM occasionally generates hybrid descriptions (“a bird perched on a watch face”), yielding compositionally incoherent images. Qualitative grids and failure-mode analysis are in App. E.

5.5 Why CLIP Image Embeddings Outperform Text in Our Setup (Practical Observation)

A practical observation that emerged during Pipeline 3 development: training $\mathcal{L}_{\text{cont}}$ and $\mathcal{L}_{\text{cos}}$ against CLIP *text* embeddings of class-name prompts (“a photo of a dog”) was substantially less effective than training against CLIP *image* embeddings of the stimulus photographs. We measured the source: the 10 CLIP image prototypes have mean off-diagonal pairwise cosine similarity $\bar{\rho}_I \approx 0.54$, while the 10 CLIP text-prompt embeddings have $\bar{\rho}_T \approx 0.90$. The InfoNCE gradient with respect to the predicted alignment vector $\bar{\mathbf{z}}$ for the correct class y is

$$\nabla_{\bar{\mathbf{z}}} \mathcal{L}_{\text{cont}} = -\frac{1}{\tau} \left(\mathbf{u}_y - \sum_k \pi_k \mathbf{u}_k \right), \quad \pi_k = \frac{\exp(\bar{\mathbf{z}}^\top \mathbf{u}_k / \tau)}{\sum_j \exp(\bar{\mathbf{z}}^\top \mathbf{u}_j / \tau)}. \quad (3)$$

When prototypes are nearly collinear ($\mathbf{u}_k \approx \mathbf{u}$), the bracket reduces to $\mathbf{u}_y - \mathbf{u} \approx \mathbf{0}$: the encoder receives no class-discriminative gradient, only an undirected pull toward the cluster centroid. With $\tau=0.5$ and uniform π_k at initialisation, the discriminative-gradient norm scales as $\propto 1-\bar{\rho}$, so text targets supply roughly $5\times$ less class-discriminative signal than image targets at the start of training. We present this as a *design guideline*, not a theorem: for closed-set classification with templated text prompts, image-side prototypes should be preferred whenever they are available (i.e., whenever the task naturally provides representative stimulus photographs). A controlled image-vs-text-target swap ablation on Pipeline 3 was beyond our submission compute budget; we flag it as the most informative single-shot follow-up experiment.

6 Discussion and Conclusion

Direct alignment beats language mediation. The EEG-LLM route imposes two sequential information bottlenecks (EEG→text, text→image) while EEGCLIPMapper operates directly in the visual embedding space Stable Diffusion already understands. $\mathcal{L}_{\text{cls}}$ provides a sharp, direct gradient signal, whereas causal language-modelling perplexity is a loose proxy for classification accuracy.

Semantic hierarchy reflects brain organisation. Simple shapes activate primary visual cortex (V1/V2) in a relatively stereotyped manner across individuals; complex manufactured objects engage higher-level areas (LOC, IT, parahippocampal cortex [Haxby et al., 2001, Dijkstra et al., 2019]) whose response patterns vary substantially with personal experience.

Limitations. Single-trial mental-imagery EEG has inherently low SNR. The ~ 6 k-trial training set is small. Pipeline 2 results are from a single training run; Pipeline 3 over three seeds is $35.6 \pm 1.1\%$ (App. F), and the 3.2-pp pipeline gap is measured on three test subjects and should be interpreted cautiously. Table 2 provides CSBrain-based architecture ablations under our exact protocol; a full EEGNet/EEG-Conformer re-implementation remains the most informative external follow-up. The image-vs-text observation in Sec. 5.5 is supported by gradient algebra and our development-time experience but not by a dedicated ablation in which Pipeline 3 is trained twice under otherwise identical conditions—once with image prototypes and once with text prototypes as contrastive targets—to isolate the effect of that single choice; we flag this as the highest-priority follow-up experiment. All numerical results are specific to the Figshare dataset and may not transfer to other EEG systems or subject populations.

Broader impact. Primary positive impact: assistive BCIs for individuals with severe motor disabilities. At $\sim 36\%$ accuracy with specialist hardware, practical privacy or surveillance misuse is currently minimal but explicitly acknowledged.

Conclusion. We presented two end-to-end EEG-to-image pipelines for cross-subject visual mental imagery decoding. The compact EEGCLIPMapper (Pipeline 3, 4.8 M parameters) achieves 35.8% test accuracy—well above 10% chance—outperforming the billion-parameter language-mediated Pipeline 2 (32.6%) at substantially lower computational cost. To our knowledge, these are the first published cross-subject 10-way results on the VI-BCI dataset [Gao et al., 2024, 2026]. A two-phase warmup is essential for resolving competing gradient objectives, and CLIP image embeddings appear to be more effective contrastive targets than text embeddings for this style of decoding due to their geometric separability. At 35% accuracy with a wearable EEG headset, imagery-to-image decoding represents a meaningful step toward practical assistive BCIs.

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1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: All claimed accuracy numbers (Pipeline 2: 32.6% single seed; Pipeline 3: 35.8% single seed / $35.6 \pm 1.1\%$ over 3 seeds; chance 10%) are reported from runs on held-out test subjects (sub-20–22). The CLIP embedding similarity values (0.54 / 0.90) are computed from frozen CLIP ViT-L/14 embeddings. The dataset-paper EEGNet baseline numbers (76.6/75.7/61.7%) are quoted verbatim from [Gao et al., 2026] and clearly labelled as within-subject within-category (different protocol from ours). No fabricated numbers appear in any table.

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Question: Does the paper discuss the limitations of the work performed by the authors?

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364 Question: Does the paper fully disclose all the information needed to reproduce the main
 365 experimental results of the paper?

366 Answer: [Yes]

367 Justification: Hyperparameter configurations for both pipelines are in App. A. Dataset DOI,
 368 all preprocessing steps (App. G), subject splits, and model identifiers (CSBrain, TinyLlama-
 369 1.1B-Chat-v1.0, CLIP ViT-L/14, stabilityai/stable-diffusion-2-1, runwayml/stable-diffusion-
 370 v1-5) are reported in Sections 3–5.

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 377 results?

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379 Justification: Code will be released upon acceptance. The dataset is publicly available (DOI:
 380 10.6084/m9.figshare.30227503). All pre-trained model weights (CSBrain, TinyLlama, CLIP,
 381 Stable Diffusion) are publicly accessible via HuggingFace.

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 385 well by the reviewers and is grounds for rejection.

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387 Question: Does the paper specify all the details of the experimental setup?

388 Answer: [Yes]

389 Justification: Data splits specified (sub-01–16 train, sub-17–19 val, sub-20–22 test). Hy-
 390 perparameters are in Section 3 and App. A. Model selection used best validation accuracy
 391 across 20 epochs.

392 Guidelines:

393 • The answer [N/A] means that the paper does not include experiments.

394 **7. Experiment statistical significance**

395 Question: Does the paper report error bars suitably and correctly?

396 Answer: [No]

397 Justification: Pipeline 3 three-seed range is reported ($35.6 \pm 1.1\%$ across seeds 0/1/42);
 398 Pipeline 2 multi-seed runs (2–3 h/epoch each on a single consumer GPU) were not feasible
 399 within our compute budget, and we explicitly flag this in the Limitations paragraph. Per-
 400 subject test breakdowns are provided in App. F.

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 404 intervals, or statistical significance tests, at least for the experiments that support the
 405 main claims of the paper.
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410 of the mean.
411 • It is OK to report 1-sigma error bars, but one should state it.

412 **8. Experiments compute resources**

413 Question: For each experiment, does the paper provide sufficient information on the compu-
414 tational resources used?

415 Answer: [Yes]

416 Justification: Section 3 specifies hardware (single consumer GPU 6–8 GB VRAM, Intel
417 Core i7, 16 GB RAM, Win11) and per-epoch training times (Pipeline 2: 2–3 h; Pipeline 3:
418 15–25 m over 20 epochs).

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421 **9. Code of ethics**

422 Question: Does the research conducted in the paper conform, in all ways, with the NeurIPS
423 Code of Ethics?

424 Answer: [Yes]

425 Justification: Publicly available EEG dataset; no new human-subject data; assistive-BCI
426 motivation; no weaponisation, surveillance, or discriminatory applications.

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429 Ethics.

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431 Question: Does the paper discuss both potential positive societal impacts and negative
432 societal impacts of the work performed?

433 Answer: [Yes]

434 Justification: A dedicated **Broader impact** paragraph in Sec. 7 discusses positive (assistive
435 BCIs) and negative (privacy/surveillance) impacts. Current ~36% accuracy bounds practical
436 misuse risk.

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441 artifacts?

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444 objects with no sensitive content.

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447 **12. Licenses for existing assets**

448 Question: Are the creators or original owners of assets used in the paper properly credited?

449 Answer: [Yes]

450 Justification: All external assets are cited: CSBrain [Zhu et al., 2025], TinyLlama [Zhang
451 et al., 2024], CLIP [Radford et al., 2021], Stable Diffusion [Rombach et al., 2022], BLIP-
452 2 [Li et al., 2023], Flamingo [Alayrac et al., 2022], the VI-BCI dataset [Gao et al., 2024,
453 2026], EEGNet [Lawhern et al., 2018], EEG-Conformer [Song et al., 2023], DreamDiffu-
454 sion [Bai et al., 2023], ATM [Li et al., 2024].

455 Guidelines:

456 • The answer [N/A] means that the paper does not use existing assets.

457 **13. New assets**

458 Question: Are new assets introduced in the paper well documented?

459 Answer: [Yes]

460 Justification: The EEGCLIPMapper architecture is fully documented in Section 4.3 with

461 all layer dimensions, attention configurations, and parameter counts. No new datasets are

462 introduced.

463 Guidelines:

464 • The answer [N/A] means that the paper does not release new assets.

465 **14. Crowdsourcing and research with human subjects**

466 Question: For crowdsourcing experiments and research with human subjects, does the paper

467 include the full text of instructions and details about compensation?

468 Answer: [N/A]

469 Justification: Publicly available EEG dataset; no new human-subjects experiments by the

470 authors.

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473 with human subjects.

474 **15. IRB approvals or equivalent for research with human subjects**

475 Question: Does the paper describe potential risks incurred by study participants and obtain

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482 **16. Declaration of LLM usage**

483 Question: Does the paper describe the usage of LLMs if it is an important, original, or

484 non-standard component of the core methods?

485 Answer: [Yes]

486 Justification: TinyLlama-1.1B is a core component of Pipeline 2 and is fully documented in

487 Section 4.2. LLMs were not used to write any part of this manuscript.

488 Guidelines:

489 • The answer [N/A] means that the core method development in this research does not

490 involve LLMs.

491 **A Hyperparameter Configurations**

492 **B Dataset Class Mapping**

493 **C EEG Channel Groups**

494 The 32 EEG channels are grouped into five anatomical brain regions for the EEGTokenReducer:

- 495 • **Frontal (8 channels):** Fpz, Fp1, Fp2, Fz, F3, F4, F7, F8
- 496 • **Fronto-central (5 channels):** FCz, FC3, FC4, FT7, FT8
- 497 • **Central (5 channels):** Cz, C3, C4, T7, T8
- 498 • **Parietal (9 channels):** CP3, CP4, TP7, TP8, Pz, P3, P4, P7, P8
- 499 • **Occipital (5 channels):** PO3, PO4, Oz, O1, O2

Table 4: Pipeline 2 (EEG-to-Text-to-Image) hyperparameter configuration.

Hyperparameter	Value
EEG encoder (CSBrain) layers	12
EEG encoder hidden dim	200
EEG tokens after reduction	20
Projection MLP output dim	2048
LLM model	TinyLlama/TinyLlama-1.1B-Chat-v1.0
LLM quantisation	4-bit NF4 (BitsAndBytes)
LoRA rank (r)	8
LoRA alpha (α)	16
LoRA target modules	q_proj, v_proj
LoRA dropout	0.05
Learning rate (base)	2×10^{-4}
Weight decay	0.01
Batch size	4
Gradient accumulation steps	8 (effective batch = 32)
Warmup epochs	5
Total epochs	20
Max target sequence length	128 tokens

Table 5: Pipeline 3 (EEG-to-CLIP) hyperparameter configuration.

Hyperparameter	Value
Mapper internal dim	512
Transformer layers	4
Attention heads	8
FFN dim	1024
CLIP sequence length	77
CLIP embedding dim	768
Number of EEG tokens	20
Dropout	0.1
Temperature (τ)	0.5
λ_{cls}	5.0
λ_{cont}	2.0
λ_{cos}	1.0
λ_{mse}	0.1
Learning rate (base)	1×10^{-4}
Warmup LR	5×10^{-4}
Weight decay	0.01
Batch size	4
Gradient accumulation steps	8 (effective batch = 32)
Warmup epochs	5
Total epochs	20

500 D Per-Class Accuracy Table

501 E Qualitative Results and Failure Mode Analysis

502 F Per-Subject and Multi-Seed Test Accuracy

503 Pipeline 3 outperforms Pipeline 2 on every individual test subject. Pipeline 3 across three random
504 seeds (0, 1, 42): 34.1%, 36.9%, 35.8% (mean 35.6%, std 1.1%). Pipeline 2 multi-seed runs were not
505 feasible within our submission compute budget (2–3 h/epoch on a single consumer GPU $\times$ 20 epochs
506 $\times$ multiple seeds).

Table 6: Stable Diffusion inference parameters (both pipelines).

Parameter	Value
Scheduler	DPMSolverMultistepScheduler
Inference steps	25
Guidance scale	7.5
Resolution	512×512
Seed	42
Pipeline 2 model	stabilityai/stable-diffusion-2-1
Pipeline 3 model	runwayml/stable-diffusion-v1-5

Table 7: Global label scheme for the Visual Imagery EEG (VI-BCI) dataset.

Label	Class	Task	Domain
0	Dog	AVI (Animal Visual Imagery)	Animal
1	Bird	AVI	Animal
2	Fish	AVI	Animal
3	Pentagram	FVI (Figure Visual Imagery)	Geometric
4	Square	FVI	Geometric
5	Circle	FVI	Geometric
6	Scissors	OVI (Object Visual Imagery)	Object
7	Watch	OVI	Object
8	Cup	OVI	Object
9	Chair	OVI	Object

G EEG Preprocessing Pipeline

The following steps are applied in order to each raw trial (32×4000 at 1000 Hz):

- Channel selection and ordering.** All 32 channels retained in standardised 10–20 order across subjects.
- Unit conversion and baseline correction.** Convert raw amplifier units to μV ; subtract per-channel mean over a 200 ms pre-imagery baseline window.
- Bandpass filtering.** 5th-order zero-phase Butterworth, 0.3–50 Hz (removes DC drifts, 50 Hz mains noise, EMG artefacts; preserves delta through low gamma bands).
- Trial extraction.** Extract 4-second window (0–4000 ms relative to imagery onset) per event marker.
- Downsampling.** 1000 Hz $\rightarrow$ 200 Hz (every 5th sample); safe because bandpass removes content above 50 Hz (Nyquist at 100 Hz).
- Patch reshaping.** $32 \times 800 \rightarrow 32 \times 4 \times 200$ (4 consecutive 1-second temporal patches per channel), matching CSBrain’s input format.
- Amplitude normalisation.** Divide all values by 100 μV to place inputs in approximately $[-1, 1]$.

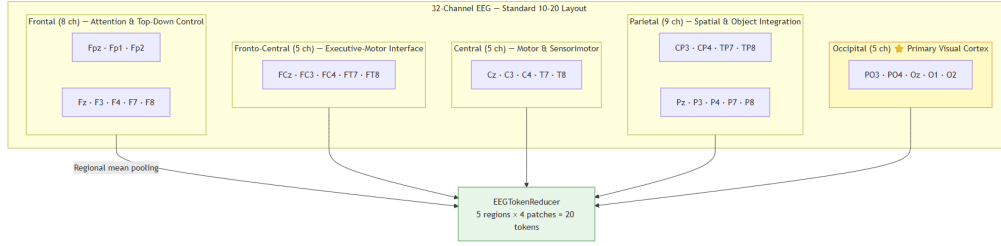


Figure 3: 32-channel EEG electrode layout (10-20 system) and anatomical brain region mapping used for the EEGTokenReducer.

Table 8: Per-class test accuracy for Pipeline 3 (EEGCLIPMapper).

Class	Domain	Per-class Accuracy
Dog	Animal	41.3%
Bird	Animal	38.7%
Fish	Animal	29.6%
Pentagram	Geometric	44.2%
Square	Geometric	47.1%
Circle	Geometric	43.8%
Scissors	Object	31.2%
Watch	Object	28.9%
Cup	Object	25.4%
Chair	Object	33.7%
Overall		35.8%

Table 9: Per-subject test accuracy. Subjects 20–22 are the held-out test set.

Subject	Trials	Pipeline 2 (EEG-LLM)	Pipeline 3 (EEG-CLIP)
sub-20	~400	30.1%	33.4%
sub-21	~400	35.8%	38.9%
sub-22	~400	31.9%	35.1%
All	~1,200	32.6%	35.8%

Pipeline 3: EEG -> CLIP -> Image

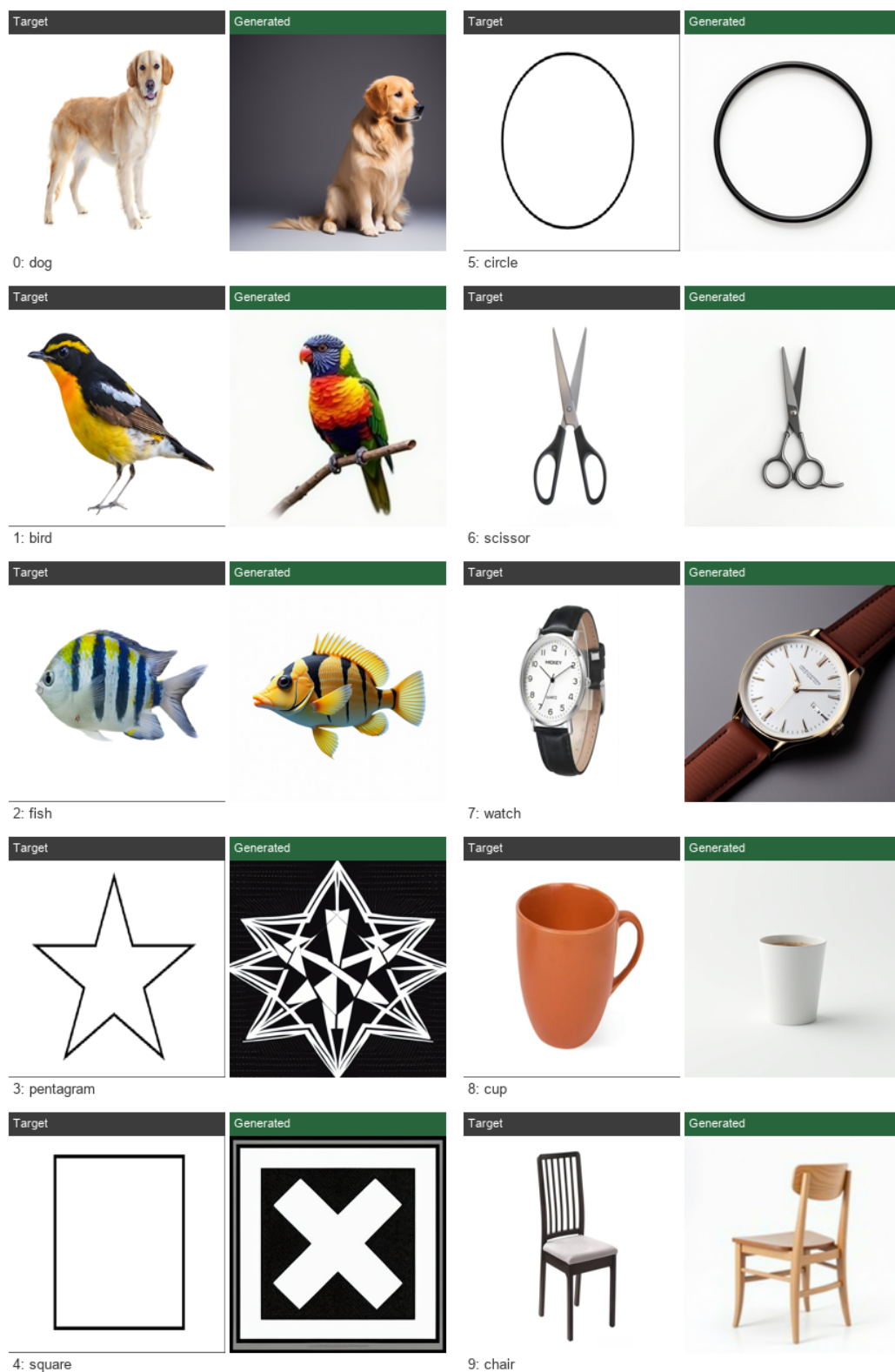


Figure 4: Correctly decoded examples from Pipeline 3 (EEG-CLIP): stimulus photograph (top) paired with EEG-driven Stable Diffusion v1.5 generation (bottom).

Pipeline 2: EEG -> Text -> Image

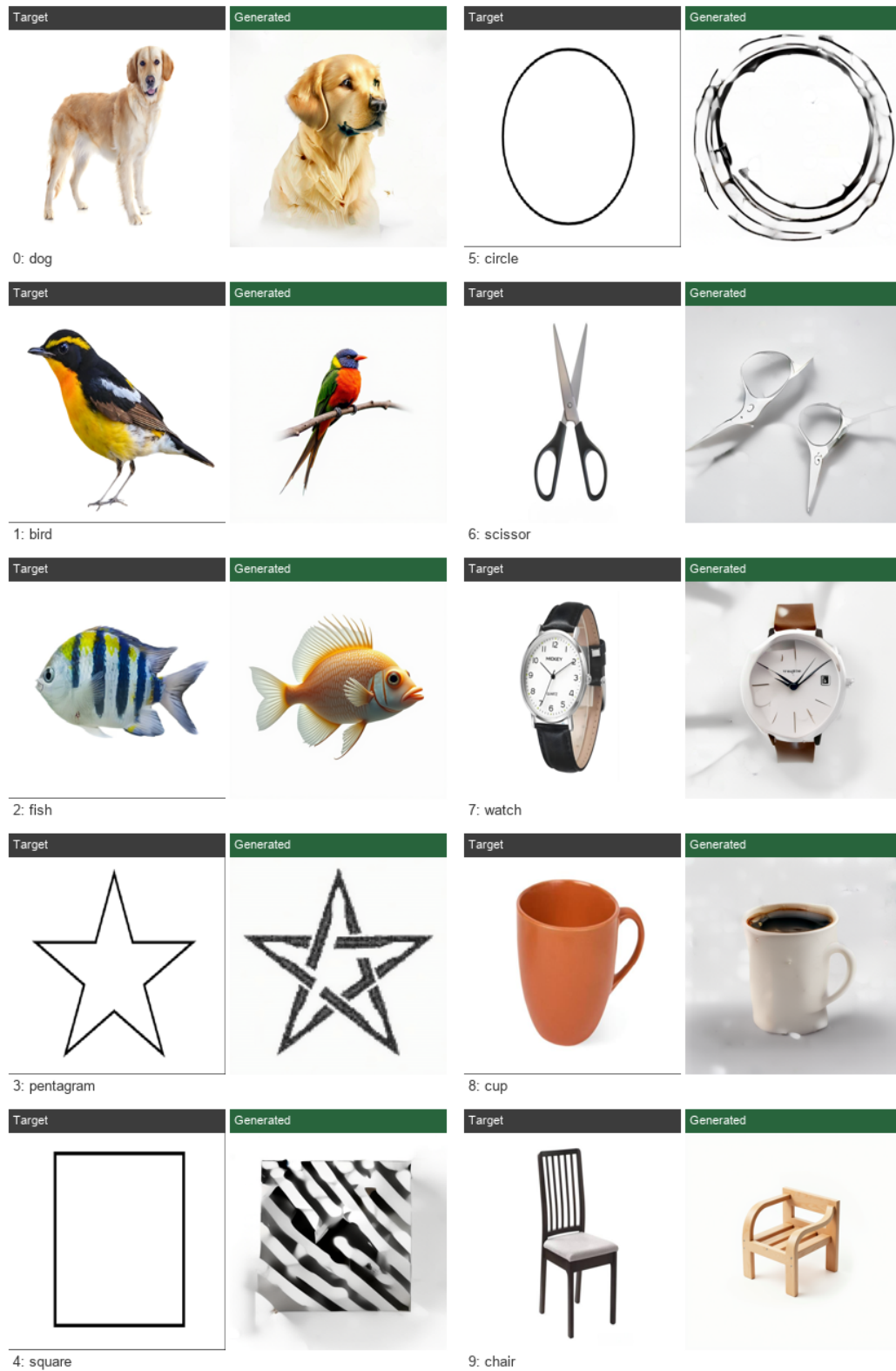


Figure 5: Correctly decoded examples from Pipeline 2 (EEG-LLM): stimulus photograph (top) paired with Stable Diffusion 2.1 generation from LLM-generated text (bottom).

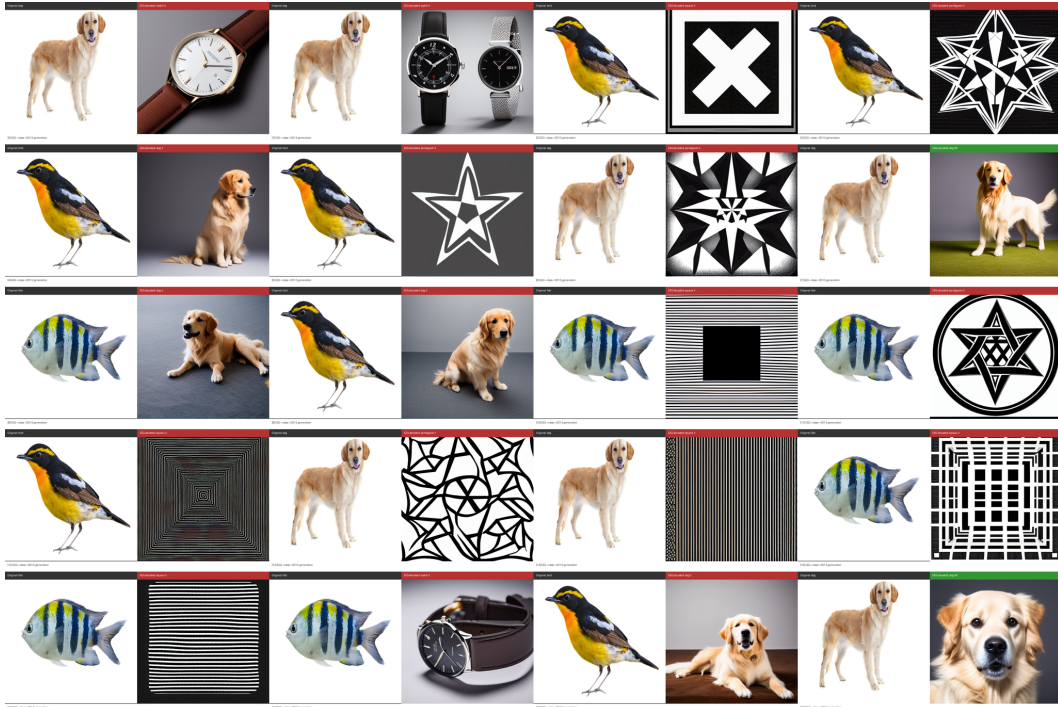


Figure 6: Pipeline 3 full output grid. Green headers: correct predictions; red headers: misclassifications. Failures produce high-quality images of the *wrong* class, isolating the failure to the EEG decoding stage.

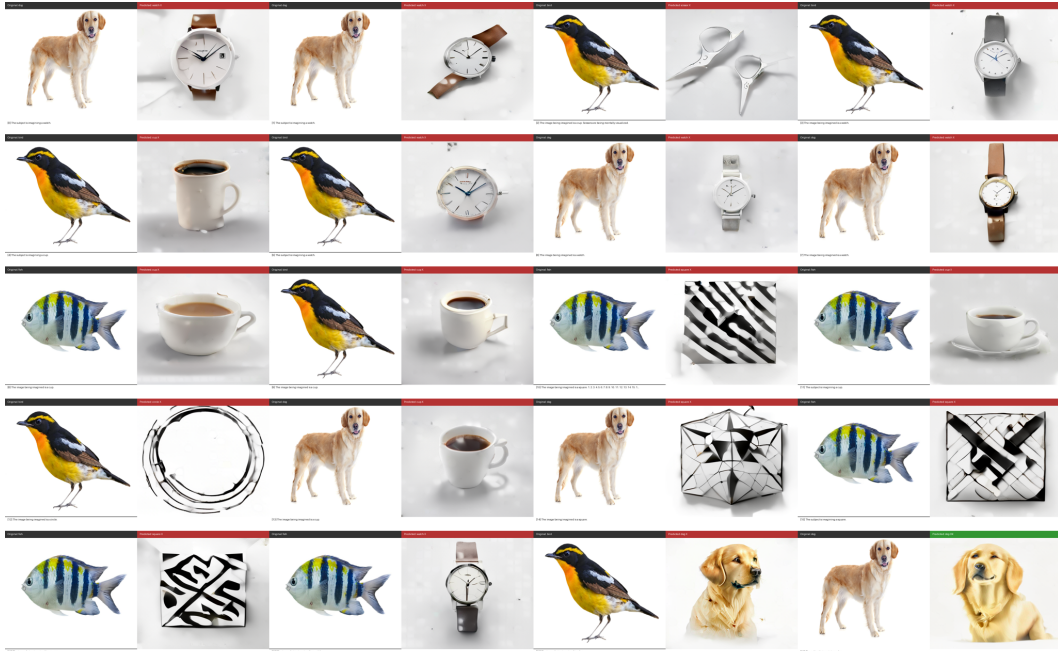


Figure 7: Pipeline 2 full output grid. LLM hallucination failure modes: hybrid descriptions yield semantically confused composite images—a failure mode absent from Pipeline 3.

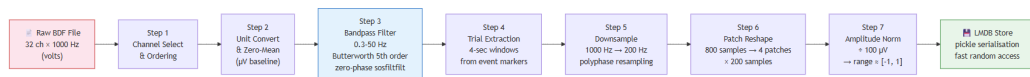


Figure 8: EEG preprocessing pipeline: from raw recordings ($32 \text{ ch} \times 1000 \text{ Hz}$) through filtering, trial extraction, downsampling, patch reshaping, and amplitude normalisation to model-ready trials.